

Example 1

Input:

```
Order of first polynomial: 2
Enter polynomial: 1 2 3 4
Order of second polynomial: 4
Enter polynomial: 5 6 7 8 9 10
```

Output:

```
First polynomial:  $4x^2 + 3x + 2 = 1$ 
Second polynomial:  $10x^4 + 9x^3 + 8x^2 + 7x + 6 = 5$ 
Sum of polynomials:  $10x^4 + 9x^3 + 12x^2 + 10x + 8 = 6$ 
Difference of polynomials:  $10x^4 + 9x^3 + 4x^2 + 4x + 4 = 4$ 
```

Example 2

Input:

```
Order of first polynomial: 3
Enter polynomial: 2 3 4 5 6
Order of second polynomial: 3
Enter polynomial: 7 8 9 10 11
```

Output:

```
First polynomial:  $6x^3 + 5x^2 + 4x + 3 = 2$ 
Second polynomial:  $11x^3 + 10x^2 + 9x + 8 = 7$ 
Sum of polynomials:  $17x^3 + 15x^2 + 13x + 11 = 9$ 
Difference of polynomials:  $5x^3 + 5x^2 + 5x + 5 = 5$ 
```

Example 3

Input:

```
Order of first polynomial: 1
Enter polynomial: 3 4 5
Order of second polynomial: 2
Enter polynomial: 6 7 8 9
```

Output:

```
First polynomial:  $5x + 4 = 3$ 
Second polynomial:  $9x^2 + 8x + 7 = 6$ 
Sum of polynomials:  $9x^2 + 13x + 11 = 9$ 
Difference of polynomials:  $9x^2 + 3x + 3 = 3$ 
```

Example 4

Input:

```
Order of first polynomial: 2
Enter polynomial: 4 5 6 7
Order of second polynomial: 2
Enter polynomial: 8 9 10 11
```

Output:

```
First polynomial:  $7x^2 + 6x + 5 = 4$ 
Second polynomial:  $11x^2 + 10x + 9 = 8$ 
Sum of polynomials:  $18x^2 + 16x + 14 = 12$ 
Difference of polynomials:  $4x^2 + 4x + 4 = 4$ 
```

Example 5

Input:

```
Order of first polynomial: 3
Enter polynomial: 2 3 4 5 6
Order of second polynomial: 4
Enter polynomial: 7 8 9 10 11 12
```

Output:

```
First polynomial:  $6x^3 + 5x^2 + 4x + 3 = 2$ 
Second polynomial:  $12x^4 + 11x^3 + 10x^2 + 9x + 8 = 7$ 
Sum of polynomials:  $12x^4 + 17x^3 + 15x^2 + 13x + 11 = 9$ 
Difference of polynomials:  $12x^4 + 5x^3 + 5x^2 + 5x + 5 = 5$ 
```

Example 6

Input:

```
Order of first polynomial: 1
Enter polynomial: 9 8 7
Order of second polynomial: 1
Enter polynomial: 6 5 4
```

Output:

```
First polynomial:  $7x + 8 = 9$ 
Second polynomial:  $4x + 5 = 6$ 
Sum of polynomials:  $11x + 13 = 15$ 
Difference of polynomials:  $-3x - 3 = -3$ 
```

Example 7

Input:

```
Order of first polynomial: 2
Enter polynomial: 3 5 7 9
Order of second polynomial: 3
Enter polynomial: 4 6 8 10 12
```

Output:

```
First polynomial:  $9x^2 + 7x + 5 = 3$ 
Second polynomial:  $12x^3 + 10x^2 + 8x + 6 = 4$ 
Sum of polynomials:  $12x^3 + 19x^2 + 15x + 11 = 7$ 
Difference of polynomials:  $12x^3 + x^2 + x + 1 = 1$ 
```

Example 8

Input:

```
Order of first polynomial: 4
Enter polynomial: 5 6 7 8 9 10
Order of second polynomial: 4
Enter polynomial: 11 12 13 14 15 16
```

Output:

```
First polynomial:  $10x^4 + 9x^3 + 8x^2 + 7x + 6 = 5$ 
Second polynomial:  $16x^4 + 15x^3 + 14x^2 + 13x + 12 = 11$ 
Sum of polynomials:  $26x^4 + 24x^3 + 22x^2 + 20x + 18 = 16$ 
Difference of polynomials:  $6x^4 + 6x^3 + 6x^2 + 6x + 6 = 6$ 
```

Example 9

Input:

```
Order of first polynomial: 3
Enter polynomial: 7 8 9 10 11
Order of second polynomial: 5
Enter polynomial: 12 13 14 15 16 17 18
```

Output:

```
First polynomial:  $11x^3 + 10x^2 + 9x + 8 = 7$ 
Second polynomial:  $18x^5 + 17x^4 + 16x^3 + 15x^2 + 14x + 13 = 12$ 
Sum of polynomials:  $18x^5 + 17x^4 + 27x^3 + 25x^2 + 23x + 21 = 19$ 
Difference of polynomials:  $18x^5 + 17x^4 + 5x^3 + 5x^2 + 5x + 5 = 5$ 
```

Example 10

Input:

```
Order of first polynomial: 2
Enter polynomial: 8 9 10 11
Order of second polynomial: 1
Enter polynomial: 12 13 14
```

Output:

```
First polynomial:  $11x^2 + 10x + 9 = 8$ 
Second polynomial:  $14x + 13 = 12$ 
Sum of polynomials:  $11x^2 + 24x + 22 = 20$ 
Difference of polynomials:  $-11x^2 + 4x + 4 = 4$ 
```

Example 11

Input:

```
Order of first polynomial: 3
Enter polynomial: 3 4 5 6 7
Order of second polynomial: 3
Enter polynomial: 8 9 10 11 12
```

Output:

```
First polynomial:  $7x^3 + 6x^2 + 5x + 4 = 3$ 
Second polynomial:  $12x^3 + 11x^2 + 10x + 9 = 8$ 
Sum of polynomials:  $19x^3 + 17x^2 + 15x + 13 = 11$ 
Difference of polynomials:  $5x^3 + 5x^2 + 5x + 5 = 5$ 
```

Example 12

Input:

```
Order of first polynomial: 1
Enter polynomial: 2 3 4
Order of second polynomial: 1
Enter polynomial: 5 6 7
```

Output:

```
First polynomial:  $4x + 3 = 2$ 
Second polynomial:  $7x + 6 = 5$ 
Sum of polynomials:  $11x + 9 = 7$ 
Difference of polynomials:  $3x + 3 = 3$ 
```

Example 13

Input:

```
Order of first polynomial: 4
Enter polynomial: 6 7 8 9 10 11
Order of second polynomial: 5
Enter polynomial: 12 13 14 15 16 17 18
```

Output:

```
First polynomial:  $11x^4 + 10x^3 + 9x^2 + 8x + 7 = 6$ 
Second polynomial:  $18x^5 + 17x^4 + 16x^3 + 15x^2 + 14x + 13 = 12$ 
Sum of polynomials:  $18x^5 + 28x^4 + 26x^3 + 24x^2 + 22x + 20 = 18$ 
Difference of polynomials:  $18x^5 + 6x^4 + 6x^3 + 6x^2 + 6x + 6 = 6$ 
```

Example 14

Input:

```
Order of first polynomial: 2
Enter polynomial: 7 8 9 10
Order of second polynomial: 2
Enter polynomial: 11 12 13 14
```

Output:

```
First polynomial:  $10x^2 + 9x + 8 = 7$ 
Second polynomial:  $14x^2 + 13x + 12 = 11$ 
Sum of polynomials:  $24x^2 + 22x + 20 = 18$ 
Difference of polynomials:  $4x^2 + 4x + 4 = 4$ 
```

Example 15

Input:

```
Order of first polynomial: 3
Enter polynomial: 8 9 10 11 12
Order of second polynomial: 3
Enter polynomial: 13 14 15 16 17
```

Output:

```
First polynomial:  $12x^3 + 11x^2 + 10x + 9 = 8$ 
Second polynomial:  $17x^3 + 16x^2 + 15x + 14 = 13$ 
Sum of polynomials:  $29x^3 + 27x^2 + 25x + 23 = 21$ 
Difference of polynomials:  $5x^3 + 5x^2 + 5x + 5 = 5$ 
```

Example 16

Input:

```
Order of first polynomial: 2
Enter polynomial: 15 12 11 10
Order of second polynomial: 2
Enter polynomial: 20 18 16 14
```

Output:

```
First polynomial:  $10x^2 + 11x + 12 = 15$ 
Second polynomial:  $14x^2 + 16x + 18 = 20$ 
Sum of polynomials:  $24x^2 + 27x + 30 = 35$ 
Difference of polynomials:  $4x^2 + 5x + 6 = 5$ 
```

Example 17

Input:

```
Order of first polynomial: 1
Enter polynomial: 5 6 7
Order of second polynomial: 2
Enter polynomial: 8 9 10 11
```

Output:

```
First polynomial:  $7x + 6 = 5$ 
Second polynomial:  $11x^2 + 10x + 9 = 8$ 
Sum of polynomials:  $11x^2 + 17x + 15 = 13$ 
Difference of polynomials:  $11x^2 + 3x + 3 = 3$ 
```

Example 18

Input:

```
Order of first polynomial: 3
Enter polynomial: 12 13 14 15 16
Order of second polynomial: 3
Enter polynomial: 17 18 19 20 21
```

Output:

```
First polynomial:  $16x^3 + 15x^2 + 14x + 13 = 12$ 
Second polynomial:  $21x^3 + 20x^2 + 19x + 18 = 17$ 
Sum of polynomials:  $37x^3 + 35x^2 + 33x + 31 = 29$ 
Difference of polynomials:  $5x^3 + 5x^2 + 5x + 5 = 5$ 
```

Example 19

Input:

```
Order of first polynomial: 2
Enter polynomial: 22 23 24 25
Order of second polynomial: 4
Enter polynomial: 26 27 28 29 30 31
```

Output:

```
First polynomial:  $25x^2 + 24x + 23 = 22$ 
Second polynomial:  $31x^4 + 30x^3 + 29x^2 + 28x + 27 = 26$ 
Sum of polynomials:  $31x^4 + 30x^3 + 54x^2 + 52x + 50 = 48$ 
Difference of polynomials:  $31x^4 + 30x^3 + 4x^2 + 4x + 4 = 4$ 
```

Example 20

Input:

```
Order of first polynomial: 3
Enter polynomial: 32 33 34 35 36
Order of second polynomial: 1
Enter polynomial: 37 38 39
```

Output:

```
First polynomial:  $36x^3 + 35x^2 + 34x + 33 = 32$ 
Second polynomial:  $39x + 38 = 37$ 
Sum of polynomials:  $36x^3 + 35x^2 + 73x + 71 = 69$ 
Difference of polynomials:  $-36x^3 - 35x^2 + 5x + 5 = 5$ 
```

Example 21

Input:

```
Order of first polynomial: 2
Enter polynomial: 0 0 3 2
Order of second polynomial: 4
Enter polynomial: 8 0 4 0 0 3
```

Output:

```
First polynomial:  $2x^2 + 3x = 0$ 
Second polynomial:  $3x^4 + 4x = 8$ 
Sum of polynomials:  $3x^4 + 2x^2 + 7x = 8$ 
Difference of polynomials:  $3x^4 - 2x^2 + x = 8$ 
```

Example 22

Input:

```
Order of first polynomial: 3
Enter polynomial: 5 0 6 0 7
Order of second polynomial: 3
Enter polynomial: 2 0 3 0 4
```

Output:

```
First polynomial:  $7x^3 + 6x = 5$ 
Second polynomial:  $4x^3 + 3x = 2$ 
Sum of polynomials:  $11x^3 + 9x = 7$ 
Difference of polynomials:  $-3x^3 - 3x = -3$ 
```

Example 23

Input:

```
Order of first polynomial: 2
Enter polynomial: 10 5 0 8
Order of second polynomial: 2
Enter polynomial: 3 0 2 9
```

Output:

```
First polynomial:  $8x^2 + 5 = 10$ 
Second polynomial:  $9x^2 + 2x = 3$ 
Sum of polynomials:  $17x^2 + 2x + 5 = 13$ 
Difference of polynomials:  $x^2 + 2x - 5 = -7$ 
```

Example 24

Input:

```
Order of first polynomial: 4
Enter polynomial: 0 1 0 2 0 3
Order of second polynomial: 4
Enter polynomial: 5 0 6 0 7 0
```

Output:

```
First polynomial:  $3x^4 + 2x^2 + 1 = 0$ 
Second polynomial:  $7x^3 + 6x = 5$ 
Sum of polynomials:  $3x^4 + 7x^3 + 2x^2 + 6x + 1 = 5$ 
Difference of polynomials:  $-3x^4 - 7x^3 + 2x^2 + 6x - 1 = 5$ 
```

Example 25

Input:

```
Order of first polynomial: 3
Enter polynomial: 9 8 7 6 5
Order of second polynomial: 3
Enter polynomial: 4 3 2 1 0
```

Output:

```
First polynomial:  $5x^3 + 6x^2 + 7x + 8 = 9$ 
Second polynomial:  $x^3 + 2x^2 + 3x = 4$ 
Sum of polynomials:  $5x^3 + 8x^2 + 10x + 11 = 13$ 
Difference of polynomials:  $-5x^3 - 4x^2 - 4x - 4 = -5$ 
```